ADS-B and ADS-C Trajectory Fusion for Gap-Filling and Route Analytics

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DATA SOURCE

Flight surveillance data was obtained from the OpenSky Network Trino database (trino.opensky-network.org), querying three tables: minio.osky.state_vectors_data4 (ADS-B position reports), minio.osky.adsc (ADS-C satellite waypoints), and minio.osky.flights_data4 (flight metadata). The study region covers the North Atlantic / Shanwick oceanic control area (35–70°N, 65°W–10°E) over the period July 2023 to August 2025, yielding 1,704 validated NAT crossing segments after cleaning and geographic filtering. The dataset is split by aircraft identity (ICAO24) into 1,094 training, 189 validation, and 240 test flights to prevent data leakage.

APPROACH

Over the North Atlantic, ADS-B ground coverage ceases beyond approximately 15°W, leaving controllers with trajectory gaps averaging 239 minutes. ADS-C satellite reports partially fill this void but arrive only every 10–40 minutes — too sparse for accurate position reconstruction. This project fuses both sources: dense ADS-B tracks are used as context to learn flight dynamics, while sparse ADS-C waypoints serve as reconstruction targets and ground truth.

A GRU encoder-decoder is trained to predict residual corrections on top of a great-circle interpolation baseline. The encoder reads 64 minutes of ADS-B before the gap and 32 minutes after; a sequential GRU decoder then predicts position offsets at each ADS-C waypoint timestamp. This residual formulation simplifies the learning problem — the model only needs to learn jet-stream deviations from the geodesic arc, not absolute positions.

SUMMARY OF METHODS

- **Great-Circle Baseline**: True spherical SLERP interpolation between the last ADS-B point before the gap and the first ADS-B point after. Evaluated at ADS-C waypoint timestamps.
- **Kalman RTS Smoother**: 4-state filter tracking along-track and cross-track position and velocity. Uses ADS-B observations from both sides of the gap; offline RTS backward pass provides non-causal smoothing.
- **GRU**: Bidirectional GRU encoder with sequential GRU decoder. Trained with exact haversine loss. Simultaneously predicts latitude, longitude, and altitude. Evaluated on 240 test flights split strictly by ICAO24.

RESULTS

All methods evaluated on the same 240 held-out test flights (mean gap: 239 min):

Method	Median Error	Mean Error	P90 Error	vs Baseline	Improvement
Great-Circle Baseline	131 km	149 km	355 km	Reference	—
Kalman RTS Smoother	83 km	98 km	222 km	**−48 km**	**−37%**
GRU	**64 km**	**84 km**	**171 km**	**−67 km**	**−51%**

The GRU model reduces median position error from 131 km to 64 km — a 51% improvement over the great-circle baseline and 23% over the Kalman smoother. This demonstrates that learned jet-stream routing patterns significantly outperform both geometric interpolation and physics-based filtering for long-duration oceanic gap reconstruction. Altitude reconstruction achieves 412 ft RMSE simultaneously with position prediction.